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DYPIU - B. Tech Bio
ACB2105
Advanced Computing for Biological Systems (Pattern - 2024)
Sem 4
2024-2028 Section-Div-1

LAB NOTES : 1

Mean, Median, Mode using IRIS Dataset

Aim of the Experiment

To understand:

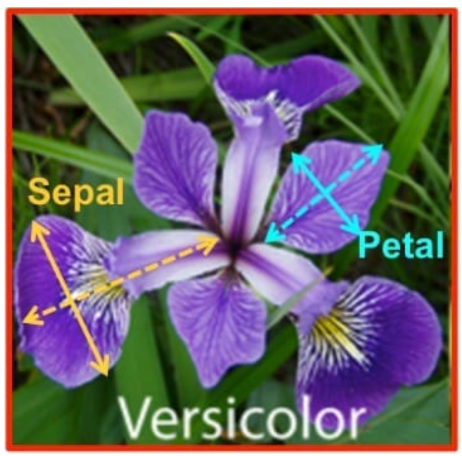
- Mean - (Sum of all values) ÷ (Total number of values)
- Median - Median is the middle number after arranging data in order.
- Mode - Mode is the value that comes most frequently.
  using the IRIS dataset, and to visualize data using graphs.

What is the IRIS Dataset?

The IRIS dataset is a very famous dataset in Data Science.
It contains details of flowers:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width
- Species (Setosa, Versicolor, Virginica)

Each row = one flower



Step 1: Import Libraries

We use libraries because they make our work easy.

- pandas → for reading and handling data
- matplotlib → for making graphs

Step 2: Load Dataset

```python
df = pd.read_csv("IRIS.csv")
```

This line reads the file and stores it in df (dataframe).

Step 3: Mean, Median, Mode

✅ Mean (Average)

Mean is the average value.

Math Formula:
Mean = (Sum of all values) ÷ (Total number of values)

Example:
Marks: 10, 20, 30
Mean = (10 + 20 + 30) ÷ 3 = 20

In code:

```python
mean_values = df.mean(numeric_only=True)
```

✅ Median (Middle Value)

Median is the middle number after arranging data in order.

Example 1 (Odd values):
10, 20, 30 → Median = 20

Example 2 (Even values):
10, 20, 30, 40
Median = (20 + 30) ÷ 2 = 25

In code:

```python
median_values = df.median(numeric_only=True)
```

✅ Mode (Most Repeated Value)

Mode is the value that comes most frequently.

Example:
2, 4, 6, 4, 8, 4
Mode = 4 (because it repeats most times)

In code:

```python
mode_values = df.mode(numeric_only=True).iloc[0]
```

Why Do We Use Graphs?

Graphs help us understand data visually instead of only numbers.

Graph 1: Histogram

Shows how data is distributed (how many values fall in a range).

```python
plt.hist(df["sepal_length"], bins=20)
```

👉 Helps to see how sepal length values are spread.

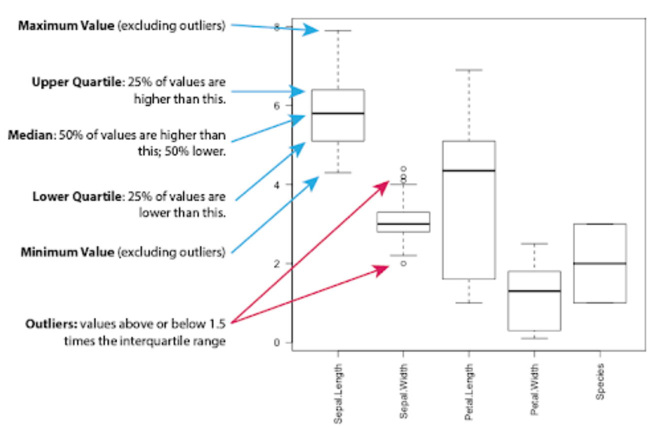
Graph 2: Boxplot

Boxplot shows:

- Minimum value
- Maximum value
- Median
- Outliers (unusual values)

```python
df.drop(columns=["species"]).boxplot()
```

👉 Helps to compare all flower features easily.

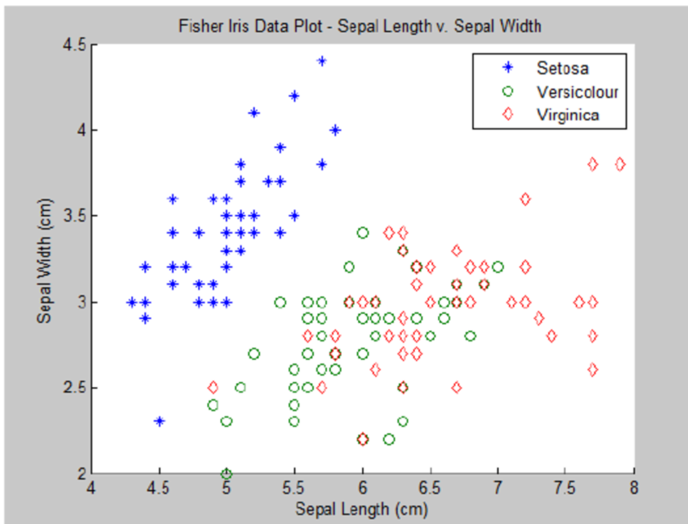


Graph 3: Scatter Plot

Shows relationship between two values.

```python
plt.scatter(df["sepal_length"], df["petal_length"])
```

👉 Helps to see if one value increases when the other increases.



Graph 4: Bar Chart

Shows total count of each category.

```python
df["species"].value_counts().plot(kind="bar")
```

👉 Shows how many flowers are there of each species.

✅ Conclusion

From this experiment, students learned:

- How to calculate Mean, Median and Mode
- How to load a dataset
- How to visualize data using:
    - Histogram
    - Boxplot
    - Scatter Plot
    - Bar Chart